

Milestone 6 Domain-Analysis Resolutions

Blockers closed per InComm direction to resolve via research, not wait

Prepared by: KSU mHealth Research Lab | Date: 2026-10-01

These are research assumptions. They are not InComm production confirmations.

Q5_issuance_ledger

- Blocker: Production issuance/redemption table not named; IncentiveRewards not implemented.
- Resolution: InComm 2026-07-20 Q4: sim_MemberRewardSimulations is a dashboard what-if tool; real issuance is a 3rd-party IncentiveRewards feed that is not implemented. Track S reconstructs gap-level assignment from the live-validated rule IsRewardEligible iff Closed. Track R generates an IncentiveRewards-shaped ledger on real members, care gaps, and catalog amounts (\$15-\$30) with Bernoulli capacity (default $p=0.45$) among gaps still open at a lagged outreach date. TreatmentSourceBasis=Research-generated IncentiveRewards mock. Never labeled InComm-issued.
- Safe claim: Track R interface is complete on a research-generated ledger. Not production issuance. Milestone 7 may consume this only as research-generated input, never as InComm-issued lift.

Q11_observation_window

- Blocker: Post-issuance observation window was unconfirmed (earlier default 90 days was a research guess).
- Resolution: Observation window = 365 days. InComm reward funds live 365 days (FundLifeSpan=365, grace 30) and InComm states every rolling period is at least a year. Track S has no IssuedAtUtc so Y is the current Closed flag. Track R mock reconstructs potential_issued_at (gap open + outreach lag) and closing-claim dates from days_since_last_qualifying_claim; Y is closure in $(t_0, t_0+365]$, administratively censored at the data cut (follow-up shorter than the window is reported, not hidden). 90/180-day windows are sensitivity only.
- Safe claim: Track R applies the 365-day window on reconstructed timestamps, not live IncentiveRewards clocks.

Q14_amount_variability

- Blocker: Whether issued amount varies within a care gap.

- Resolution: InComm's catalog (ref_ConditionRewardMappings) has Min/Default/Max per gap, and InComm's wallet configuration (stg_WalletIncentivizedActivities, 6,093 program x activity rows) shows reward amounts for the same activity vary across programs (\$1-\$100, median about \$20). Track S sim rows use the default only (no variation, elasticity NOT_AVAILABLE there). The Track R mock draws each offer's amount from the empirical per-gap pool, independent of outcome, so the dose-response estimator runs and its power is known. Real members are not linked to programs (can_Members.ProgramCode is null), so observational elasticity from InComm data is impossible; a randomized amount arm is the route.
- Safe claim: Elasticity is estimated on the research mock only (method + power), never as InComm's response curve.

diabetes_positivity

- Blocker: Diabetes cohort has an empty untreated arm in the member-broadcast contrast.
- Resolution: That empty arm is an artifact of broadcasting a member's lifetime reward count onto every gap (M4 grouped sim rows by MemberId). At gap level, every diabetes gap has both arms. A real positivity violation (every unit in a stratum treated) is handled by: (1) never extrapolating tau into the stratum, (2) reporting it NOT_IDENTIFIED with partial-identification bounds, and (3) specifying the holdout size that would restore identification (positivity table, reports/milestone6_positivity.json).
- Safe claim: No tau is reported for a stratum that fails positivity.

ckd_exclusion

- Blocker: CKD_NEPHROLOGY_VISIT is excluded from the headline.
- Resolution: Deliberate, and the reason is a platform defect, not a data gap: InComm's rule never fired because ref_CareGapDefinitions.ConditionCode='CKD' does not match can_ConditionFlags 'CHRONIC_KIDNEY_DISEASE', so der_MemberCareGapStatuses has zero CKD rows. The headline uses InComm-sourced labels only. CKD runs as a separate secondary cohort on research-computed labels (1,666 rows), reported beside the headline and never mixed in.
- Safe claim: CKD results are research-labeled and secondary.

as_of_vs_closing_claim

- Blocker: Issued-at relative to closing claim is unconfirmed.
- Resolution: ISS-008 (live extract): every can_ClinicalClaimLines ServiceFrom is on or before 2026-06-23; earliest der_RecommendationCases.CreatedAtUtc is 2026-06-25. HAR / sim

TriggeredAtUtc is after every claim. Track R mock reconstructs outreach lag from gap tenure so issued_at can precede a later closing claim; rows already closed at t0 are ineligible (real-world: do not issue after close).

- Safe claim: HAR time order stays reversed. Mock timestamps are research reconstructions.

assignment_mechanism

- Blocker: No campaign A/B or holdout named.
- Resolution: Track S assignment = InComm ML HEALTH_ACTION_REWARD eligibility (Closed $\Leftrightarrow$ eligible $\Leftrightarrow$ HAR case), so T equals Y. Track R mock assignment is capacity-constrained Bernoulli among gaps still open at t0, independent of later Closed. That is a research holdout shape, not an InComm campaign A/B.
- Safe claim: Track S T is a function of Y. Track R mock T is independent of later Closed.

incentive_rewards_holdout

- Blocker: Observational CATE cannot be identified while T is determined by Y.
- Resolution: Track R mock: (1) dated gap-level issued_at, (2) drop issue-after-close, (3) 365-day Y after potential issue, (4) untreated arm among eligible Open and later-closed gaps. Production IncentiveRewards is still absent. Mock CATE is a negative-control / pipeline result (historical Y was not caused by mock T). Milestone 7 may consume this only as research-generated input, never as InComm-issued lift.
- Safe claim: Production causal lift still requires a named IncentiveRewards feed.

who_acts

- Blocker: Care-manager vs campaign capacity unknown.
- Resolution: Targeting is a ranked list plus Treat/Hold/Suppress volume curve.
- Safe claim: No single operating threshold.